1. 分离变量.

$$\begin{cases} \dfrac{\partial^2 u}{\partial t^2} - a^2 \dfrac{\partial^2 u}{\partial x^2} = 0 & (0<x<l,\ t>0) & (1,a) \\ u|_{x=0} = 0,\ u|_{x=l} = 0 & (t \geq 0) & (1,b) \\ u|_{t=0} = \varphi(x). \quad \dfrac{\partial u}{\partial t}\Big|_{t=0} = \psi(x). & (0 \leq x \leq l) & (1,c) \end{cases}$$

考虑. $u(x,t) = X(x)\,T(t)$. 代入 $(1,a)$ 有:

(1) $$\frac{X''(x)}{X(x)} = \frac{T''(t)}{a^2 T(t)}$$

$\frac{X''(x)}{X(x)}$ 为 t 无关; $\frac{T''(t)}{a^2 T(t)}$ 为 x 无关.

即 $\forall x, t$, 上式成立. $\quad \frac{X''}{X} = \frac{T''}{a^2 T} = -\lambda$.

$$\begin{cases} X'' + \lambda X = 0 \\ T'' + a^2 \lambda T = 0. \end{cases}$$

(2) 将 $u(x,t) = X(x)\,T(t)$ 代入定解条件. $\begin{cases} u|_{x=0} = 0 \quad u|_{x=l} = 0 \\ u|_{t=0} = \varphi(x) \quad \dfrac{\partial u}{\partial t}\Big|_{t=0} = \psi(x) \end{cases}$ 有:

$$X(0)\,T(t) = X(l)\,T(t) = 0$$

$$X(x)\,T(0) = \varphi(x) \qquad X(x)\,T'(0) = \psi(x)$$ 一般地, 该式并不满足, 先不考虑

(3) 先考虑方程 $X'' + \lambda X = 0$